

检测系统研发选题之中期检查自述表

姓名	杨礼谦	学号	2022080054	Email	rqyong@gmail.com
题目	Formula 1 Racing Car Components Reliability Assessment and Integrity Monitoring System (CRAMS)				
摘要	Research Gap and Background Despite advances in sensor fusion and predictive maintenance in various industries, a dedicated solution tailored to the extreme, high-stress environment of Formula 1 racing remains elusive. F1 cars operate under intense conditions—rapid acceleration, high temperatures, and significant vibrations—that demand real-time, predictive monitoring of component integrity. Current systems mainly focus on performance optimization or generic health monitoring, leaving a gap in integrating multi-modal sensor data for reliability assessment and proactive maintenance. In F1, even the slightest fault can result in catastrophic failures, race retirements, or costly downtime. With hundreds of sensors generating vast amounts of data during races, there is a critical need for a system that not only leverages this data for immediate performance insights but also forecasts potential degradation. This combined challenge of ensuring both optimal performance and preemptive fault detection underscores the necessity for a specialized, sensor-based monitoring solution in Formula 1. Analysis of Domestic and International Research Status Foreign Research Progress Equipment Health Monitoring Technology - GE's Predix Platform: GE's Predix platform leverages IoT technology for real-time monitoring of industrial equipment. Despite its high cost and primary application in large-scale industrial scenarios, it demonstrates the potential of IoT technology in equipment health monitoring. This technology can collect real-time equipment operation data for health assessment and fault prevention. - Fraunhofer Institute's Acoustic Emission Sensors: The Fraunhofer Institute has developed acoustic emission sensors for bearing fault detection. This mature technology, which requires specialized hardware, can detect micro acoustic emission signals from within equipment to identify potential faults early, making it suitable for health monitoring of critical equipment. Application in the Racing Field - On-board Sensors in F1 Racing: F1 teams have adopted on-board sensors to monitor parameters like engine temperature and vibration in real-time, mainly for performance optimization. These sensors provide real-time data to help engineers adjust the car's settings and enhance race performance. However, this technology is primarily used for performance improvement rather than predicting the lifespan of components. Domestic Research Progress Multi-modal Fusion Technology - Huawei's Multi-modal Industrial Inspection Patent: Huawei applied for a "Multi-modal Industrial Inspection" patent in 2023, which integrates vibration and visual signals for equipment health assessment and can be migrated to the racing scenario. This technology uses multi-source data (vibration, temperature, electrical signals) collection and health index calculation framework for multi-dimensional monitoring of racing car components. - Real-time Monitoring System: East China Technology obtained a patent for a "Health Monitoring System Based on Broadband Fieldbus" in 2025, which supports high-speed data transmission and edge computing, making it suitable for the high-frequency data requirements of racing cars.				

	System Indices Performance Index: 1. Weight Constraint: The total weight of the sensors have to be less than 400g to avoid affecting the racing car's performance. 2. Latency Requirement: The data delay must not be more than 10ms. 3. Robustness and Durability: The sensors must be able to withstand high temperature and vigorous vibration, as Formula 1 cars operate in extreme conditions with high speeds and significant mechanical stress. 4. Data Accuracy and Precision: Sensors within CRAMS must provide highly accurate and precise data to detect subtle changes in component performance or emerging faults. For example, vibration sensors should have a sampling frequency of at least 10kHz to capture detailed vibration characteristics, and the infrared temperature measurement module should have a precision of $\pm 0.5^{\circ}\text{C}$. 5. Integration Capability: CRAMS should seamlessly integrate with the vehicle's existing telemetry and data analysis infrastructure. 6. Safety and Compliance: This system must comply with the mandatory safety requirements set by the FIA and relevant international standards.
参考文献目录	[1] L. Belkovic and I. Takács, "The Digitalization of Formula 1: Innovations from Car Design to Race Strategy," 2023 IEEE 21st Jubilee International Symposium on Intelligent Systems and Informatics (SISY), Pula, Croatia, 2023, pp. 000565-000570, doi: 10.1109/SISY60376.2023.10417955. [2] A. Gu, A. Andreyev, M. Terada, T. Rodgers and V. Viswanathan, "A Deep Learning Reconstruction Technique and Workflow to Enhance 3D X-ray Imaging Resolution and Speed for Electronics Package Failure Analysis," 2023 International Conference on Electronics Packaging (ICEP), Kumamoto, Japan, 2023, pp. 69-70, doi: 10.23919/ICEP58572.2023.10129701. [3] P. Karle, F. Fent, S. Huch, F. Sauerbeck and M. Lienkamp, "Multi-Modal Sensor Fusion and Object Tracking for Autonomous Racing," in IEEE Transactions on Intelligent Vehicles, vol. 8, no. 7, pp. 3871-3883, July 2023, doi: 10.1109/TIV.2023.3271624. [4] Kullu, O.; Cinar, E. A Deep-Learning-Based Multi-Modal Sensor Fusion Approach for Detection of Equipment Faults. Machines 2022, 10, 1105. https://doi.org/10.3390/machines10111105 [5] Qin Tang, Jing Liang, Fangqi Zhu, A comparative review on multi-modal sensors fusion based on deep learning, Signal Processing, Volume 213, 2023, 109165, ISSN 0165-1684, https://doi.org/10.1016/j.sigpro.2023.109165. [6] Buabeng, Albert & Simons, Anthony & Frempong, Nana Kena & Yevenyo Ziggah, Yao. (2022). Predictive Maintenance Model Based on Multisensor Data Fusion of Hybrid Fuzzy Rough Set Theory Feature Selection and Stacked Ensemble for Fault Classification. Mathematical Problems in Engineering. 2022. 10.1155/2022/4372567. [7] Md. Abdur Rahim, Md. Mustafizur Rahman, Md. Shofiqul Islam, Abu Jafar Md. Muzahid, Md. Arafatur Rahman, Devarajan Ramasamy, Deep learning-based vehicular engine health monitoring system utilising a hybrid convolutional neural network/bidirectional gated recurrent unit, Expert Systems with Applications, Volume 257, 2024, 125080, ISSN 0957-4174, https://doi.org/10.1016/j.eswa.2024.125080. [8] Khanmohamadi, M., & Guerrieri, M. (2024). Advanced Sensor Technologies in CAVs for

	Traditional and Smart Road Condition Monitoring: A Review. Sustainability, 16(19), 8336. https://doi.org/10.3390/su16198336 [9] Gültekin, Ö., Cinar, E., Özkan, K., & Yazıcı, A. (2022). Real-Time Fault Detection and Condition Monitoring for Industrial Autonomous Transfer Vehicles Utilizing Edge Artificial Intelligence. Sensors, 22(9), 3208. https://doi.org/10.3390/s22093208
--	--